

# Teaching Environmentally Conscious Game Design: Lessons and Challenges

SONIA FIZEK, TH Köln University of Applied Sciences, Germany
MIKHAIL FIADOTAU, Tallinn University, Estonia
HANNA WIRMAN, IT University of Copenhagen, Denmark
MARIA GARDA, University of Turku, Finland



A growing number of game design and development programs in higher education are beginning to incorporate environmental considerations into their curricula. This aligns with the broader push for more responsible and environmentally conscious practices in game production, consumption, and education. This article introduces two ongoing European inter-university projects: the Erasmus+ Greening Games project and the Nordplus-funded Nordic Alliance for Sustainability in Gaming network. Using them as case studies, the article reflects on the lessons learned and the challenges encountered while implementing the projects' pilot stages, as well as their implications for environmentally conscious game design education more broadly.

CCS Concepts: • **Applied computing** → **Computer games**; • **Hardware** → **Impact on the environment**; • **Social and professional topics** → **Sustainability;**

Additional Key Words and Phrases: Game design, sustainability, education, green computing

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## 1 INTRODUCTION

Videogames are embedded into our culture and daily life, both reflecting and shaping the social context and values of the times. As McKenzie Wark argues, videogames are "a key part of the shared culture from which one can begin ... the process of creating a reflective and critical approach to the times" [23].

Our relationship with nature and the environment is a case in point. In a world undergoing a climate crisis, many videogames have turned to environmental themes, offering different visions for our future as a species and a planet [1, 5]. This is a promising development considering that

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Authors' addresses: S. Fizek, TH Köln University of Applied Sciences, Schanzenstr. 28, 51063 Köln, Germany; email: sf@colognegamelab.de; M. Fiadotau, Tallinn University, Narva mnt. 29, 10120 Tallinn, Estonia; email: fiadotau@tlu.ee; H. Wirman, IT University of Copenhagen, Rued Langgaards Vej 7, 2300 Copenhagen, Denmark; email: wirman@itu.dk; M. Garda, University of Turku, Siltapuistokatu 2, 28100 Pori, Finland; email: maria.garda@utu.fi.
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gaming is a medium that reaches over 3 billion people worldwide [9], many of whom are aware of, and at least somewhat concerned about, climate change [17]. But this also has additional ramifications, since that gaming and the videogame industry have themselves been notable contributing factors to environmental issues [2].

In recent years, the growing recognition of videogames' environmental impact has brought about numerous initiatives in the game industry. In 2019, the United Nations launched its Playing for the Planet Alliance initiative, whose members (including industry giants Sony, Ubisoft, and Microsoft) seek to lead by example and commit to adopting more environmentally sustainable practices. As a grassroots-level counterpart to this, the International Game Developers Association has formed a **Special Interest Group (SIG)** on climate change, focused on reaching the broader game development community. In spring 2022, the SIG published its **Environmental Game Design Playbook (EGDP)**, combining expertise from environmental psychology and game design to "establish a common design language for discussing climate action in games" [24].

Important efforts are underway at the national and regional level as well, from Ubisoft's Play Green initiative and decarbonization plan to the German Games Industry Association's 2021 guide on environmental and climate protection. In the Nordic region, the industry-led PlayCreateGreen group has established a community-driven "Climate Handbook for Game Companies by Game Companies," breaking down the environmental impact of gaming into three stages: development and production, marketing and promotion, and gameplay.

While some environmental initiatives in the game industry have been criticized as misguided or driven chiefly by PR concerns [8], the rise of such initiatives testifies to the importance of environmental action in today's videogaming. As researchers and educators, we recognize that game design education must, too, respond to this change. It is our responsibility to promote ethical game design and sustainable gaming culture among future game professionals, who can then inspire informed collective action in the industry and the broader game development community.

Doing so, however, is no trivial task. It is not enough to rehash existing ideas and mechanics while adding cosmetic environmental elements. Designers and game development teams need to learn to use videogames consciously as a medium capable of both critical expression and representation of systemic complexity. This requires a combination of skills including basic ecoliteracy [10], which is not typically found in game design programs.

So how do we teach environmentally considerate game design? What should the students know and what should their instructors aim for? In this article, we will explore this question, providing insights and practical tips based on our experience in two ongoing projects: *Greening Games: Building Higher Education Resources for Sustainable Video Game Production, Design and Critical Game Studies* (Erasmus+ Higher Education) and *Nordic Alliance for Sustainability in Gaming* (Nordplus).

## 2 CASE STUDY 1: NORDIC ALLIANCE FOR SUSTAINABILITY IN GAMING

The *Nordic Alliance for Sustainability in Gaming* is a Nordplus-funded network involving members from six universities: Tallinn University (Estonia), the University of Turku (Finland), IT University of Copenhagen (Denmark), Uppsala University and the University of Skövde (both Sweden), and the University of Iceland. The goal of the network is to promote environmental and cultural sustainability in games education and the videogame industry. A notable feature of the network is its integration of aspects related to environmental and cultural sustainability, viewing diversity, inclusivity, and job security in the game industry as parts of the same continuum as its environmental footprint.[1]

[1]More information about the network is available on its website: http://nasg.tlu.ee.

In fall 2021, the network organized an academic seminar in Turku, Finland, dedicated to sustainability in gaming, involving game researchers, industry representatives, educators interested in the use of games, and game design teachers and students. This was followed in June 2022 by the inaugural NordEcoJam: an intensive course and international game jam at Tallinn University in Estonia, which brought together 29 student participants from the partner universities. During the 1-week event, participants were offered talks and workshops by the network members, as well as invited speakers, including game researchers (Ben Abraham), environmental scientists (Jaanus Terasmaa), and a representative of the International Game Developer Association's SIG on Climate (Paula Escuadra). The focus of the talks ranged from an overview of the environmental challenges facing the world to the issue of greenwashing and the limits of carbon offset programs to approaches to designing and developing videogames more sustainably. Against this backdrop, participants were grouped into international teams and were tasked with prototyping digital games that would in some way reflect on the issue of environmental and cultural sustainability. (There was no theme: the scope was deliberately kept broad to allow for a wider range of foci and approaches.)[2]

A key feature of the approach taken in NordEcoJam was to encourage participants to work sustainably, with the event designed to discourage crunch time and reduce the stress often associated with game jam participation [7], as well as game industry work more broadly. Since some of the participants were not game design students (e.g., the Icelandic participants were from a teacher training program), it was important to harmonize expectations and ensure that the skill sets within every team were balanced and complemented each other. This included drawing on the Icelandic participants' background in instructional design and the Finnish students' expertise in cultural production studies. To ensure participants received enough support, each team was assigned a mentor from the organizing team, with other network members providing additional feedback ad hoc.

The game jam yielded six game prototypes, including one non-digital card game, ranging from a city-building simulation with an environmental twist (after designing their futuristic city, the player is suddenly transported into the role of a duck trying to survive in the urban environment) to a post-apocalyptic narrative experience exploring the ethical ramifications of destroying the planet.

## 3 CASE STUDY 2: GREENING GAMES

*Greening Games: Building Higher Education Resources for Sustainable Video Game Production, Design and Critical Game Studies* is an interdisciplinary project conducted by research teams at four higher education institutions: TH Köln University of Applied Sciences (Germany), Breda University of Applied Sciences (The Netherlands), Charles University (Czech Republic), and the University of Turku (Finland).[3] The aim of the project is to develop, test, and disseminate teaching materials and know-how to support educators in addressing the interdisciplinary nature of green digital gaming. These materials are to be tested in selected higher education programs and eventually shared with the broader academic and teaching community via an open-access repository. The project is supported by the Erasmus+ Program of the European Union and the Federal Ministry of Education and Research in Germany.

In the summer semester of 2022, the project conducted a pilot intervention, which was integrated into the existing Collaborative Projects course (module in UK terms) within the Digital Games BA study program at TH Köln's Cologne Game Lab. During the second half of the semester, undergraduate students from three specializations (game arts, game programming, and game

[2]More information on the event is available at https://nasg.tlu.ee/nordecojam22-program/.
[3]For more information, see the project website http://greeningames.eu.

design) worked in small teams to develop prototypes for environmentally themed games. The pilot involved 39 students, who formed 10 teams, with each team designing their own game: a playable prototype with at least 5 minutes of gameplay.

The learning objective was to explore videogames as playable forms of critical ecological thought. The process drew on Mary Flanagan's concept of radical game design [12], encouraging students to make conscious design choices that supported their intended message, as well as reflect on the political and social implications of their work. To learn more about designing environmental games specifically, the students had to familiarize themselves with selected concepts from the aforementioned EGDP.

Four mentors from the Cologne Game Lab supported the students in the process, each covering a different area of expertise: game studies, game programming, game design, and game arts. In addition, three external mentors from the IGDA Climate SIG were involved: Paula Angela Escuadra, Trevin York, and Hugo Bille. They participated remotely in the game pitching session and intermediate presentations before attending the final game presentations on site in Cologne. Individual feedback was also provided via a dedicated Discord server.

The resulting prototypes were diverse in genre and concept, ranging from a story-driven platformer whose protagonist is a plastic bag traveling the world (*Hold My Hand, I'm Scared*), to a location-based mixed reality game that required players to physically travel to an information center in the Cologne area dedicated to wild bees (*Beeware*).

## 4 LESSONS LEARNED

The two interventions (NordEcoJam and the Greening Games pilot) not only produced a diverse set of innovative game prototypes but also provided much food for thought for participants and organizers alike. Below, we reflect on several key takeaways, where our experiences with the two projects align. While a more formal analysis of empirical data collected during and after the interventions will be published later, these initial takeaways are based on observations during the interventions, playtesting student games, and discussion in reflective sessions (some of which involved the participants, while others were limited to organizers).

### 4.1 Foster Conscious Game Design

Developing a game with an environmental theme or visual aesthetic may seem relatively simple at first. After all, students in game design programs are likely already equipped with the skills needed to make engaging and appealing games, regardless of specific subject. However, many students are proficient in neither the rhetorical capabilities of the medium nor environmental science. (The latter point also often applies to their instructors.)

It is thus crucial to set aside time before the game creation phase for students to understand specific environmental issues and explore the different ways in which these issues can be represented and operationalized in games. To make conscious game design decisions, choosing game mechanics and aesthetics to fulfill a specific rhetorical purpose, students would benefit from engaging with such concepts as critical play [12], procedural rhetoric [6], systemic design [22], and operational logics [20]. A good place to start from is the EGDP, in particular Part Two, which outlines the "tactics" of designing environmental games, breaking them down into such categories as mechanics and procedural rhetoric, narrative tactics, mixed reality designs, and system knowledge and simulations [24].

Another helpful task to consider prior to the design is an analysis of selected environmental games with a specific focus on their design patterns, mechanics, themes, and aesthetics. The games can be chosen both as examples of how game design can be done consciously and as anti-examples of rehashing existing mechanics and conventions uncritically—perhaps a mixture of the two can

provide the most discussion fodder. Involving subject matter experts from environmental science is a good way to complement the game design instructors' expertise and make sure the game ideas actually reflect real-life environmental processes and issues.

### 4.2 Dream Global, Design Local

Global-scale problems are difficult to design for, as they often seem overwhelming or vague and disembedded from students' lived experience. Focusing on local issues that exemplify broader environmental concerns is a way to make the design process more concrete, personally meaningful, and emotionally resonant. Many regional and municipal authorities coordinate the implementation of the UN **sustainable development goals (SDGs)** at the local level, addressing the challenges and needs of their specific communities [11]. These strategies provide a useful reference point not just for environmental action but also for game design projects, enabling them to engage with local contexts and address specific issues affecting a particular area. For example, the city of Turku in Finland has published a Voluntary Local Review on the implementation of the UN's 2030 Agenda, including an up-to-date roadmap for reaching the SDGs [13]. The document reports on achieved progress, identifies ongoing challenges, and outlines opportunities for community participation. A critical review of such documents could be a useful starting point for more locally and civically engaged game design, idea generation, and selection.

An alternative approach is to choose an existing environmental project on which sufficient and reliable data are available. For example, during the Greening Games pilot, one of the teams chose as their point of reference the isle of El Hierro in the Canary Islands, which is entirely powered by renewable energy. The students modeled the island as the setting for an augmented reality experience with the aim of highlighting an environmental success story.

### 4.3 See Beyond Simplistic and Overly Optimistic Environmental Narratives

The growing recognition of the urgency of climate change has led to changes in policy-making and business practices across the globe. Many of these changes, however, have been criticized as inadequate, misguided, or outright deceptive. The practice of corporate greenwashing–"manufactured optimism" aimed at concealing the ongoing destruction of the environment by major corporate players [3]–is a case in point. It is thus crucial that an environmentally conscious education promotes eco-literacy, including the ability to see past PR language and critically analyze environmental discourses.

Many current environmentally themed videogames present an overly optimistic take on climate change, making the issue appear solvable within the very paradigm of constant "progress and growth" that gave rise to it [5]. Other games reduce environmental processes and climate action to simple, often rehashed mechanics: the numerous recycling and waste sorting games are a prominent example, typically providing little novelty in terms of content and gameplay.

In NordEcoJam, the concept of greenwashing itself became the subject of one of the teams' games: a card-based simulation of a marketing agency brainstorming for ways to greenwash environmentally unsustainable products. While humorous in tone and emphasizing social gameplay over factual accuracy, the game helped highlight a pervasive but problematic practice rarely tackled in environmentally themed games.

### 4.4 Make Sustainability Part of the Game Creation Process

Using videogames as a persuasive medium to change players' attitudes and behavior with regard to the environment is a key consideration in game design education. However, it is equally important to consider how game consumption and production themselves affect the world. The videogame

industry, after all, has exerted a significant negative impact on the environment, with a carbon footprint "potentially greater ... than the global film industry" [2].

Moreover, the game industry has a problematic track record in terms of social and cultural sustainability, struggling throughout its history with issues such as crunch time, as well as a lack of diversity and inclusivity. These issues, too, speak to environmental sustainability. Consider, for example, the issue of environmental inequality, whereby marginalized communities are often the ones to bear the brunt of environmental pollution and climate change [18]. Such marginalized communities have often been excluded from participation in the game industry, preventing them from telling their stories and sharing their perspectives, including on environmental issues [14].

It is crucial for future game professionals to recognize these issues and adopt a sustainability-conscious mindset in regard to their own practice. It is equally crucial, then, to run game creation events (and other learning activities) in a way that supports this mindset. For example, during NordEcoJam, we made a point of limiting scheduled activities (including workshops and jamming time) to around 8 hours a day. This helped discourage crunching (as classrooms would close in the evening) and maximize the use of natural light. Reflecting on such organizational choices during the debrief is important to make sure students understand the rationale behind them.

### 4.5 Support Responsible Play

While the videogame industry is the larger contributor to climate change, the environmental impact of game consumption is also sizable and should not be disregarded. This mostly pertains to energy consumption but also to hardware and media recycling [4].

Promoting sustainable play at the end-user side can be a challenge for game designers and developers, given the industry's historical focus on high-end computational performance, short product lifecycles, and planned obsolescence. However, practices such as "green coding" (programming code that aims at minimizing energy consumption), careful consideration over the necessity of resource-hungry features (such as high frame rates and photorealistic graphics), as well as design that considers the duration and technological requirements of meaningful play experiences can contribute to a paradigm shift.

The choice of platform is of particular importance, as mobile devices and laptops typically consume less energy than high-end gaming PCs or latest generation home video game consoles [2]. Supporting older platforms is another consideration, as it does not force players to abandon their still functional devices in favor of the latest hardware, reducing hardware recycling issues. Making informed decisions about what platform both fits the game's goals and serves its developers' values is thus an important part of environmentally conscious game design.

## 5 CHALLENGES

### 5.1 Balancing Hope and Urgency

The climate crisis is, without question, an extremely high-stakes situation, not to mention one where much damage has already been done and progress has been inadequate. Further, discussing climate change, as well as other pressing environmental concerns such as the loss of biodiversity, often highlights the limited ability of individual people and grassroots communities to effect systemic change to the status quo. Research indicates that many young people already harbor a sense of pessimism around climate change [19]. Much of it is due to their perceived inability to make a difference amid the "grand narrative of environmental gloom and doom" [15]. This often translates into inaction and a sense of helplessness.

Hope–both in the fundamental possibility of change and one's own ability to contribute to it–is thus a crucial motivator for environmental action [19], including through game design [24].

However, hope rooted in denial of the issue's existence or severity is counterproductive [19], and it is equally important to communicate the urgency of the issue and the need for immediate action, including at the grassroots level.

Following the first day of NordEcoJam, several participants approached their team mentors with the same concern. After getting an overview of the environmental crisis and learning about the inadequacy of the measures currently taken to mitigate it, they were questioning whether, as game design students, there was anything meaningful they could do to help with what seemed like an insurmountable challenge. Despite reassurances that grassroots participation was key to systemic change, one of the teams ended up producing what seems a pessimistic take on the issue. In their choice-based adventure game, a team of survivors from the environmental apocalypse on Earth are seeking refugee status on another planet. The survivors must stand trial in front of a board of alien jurors and take accountability for humanity's actions on their home planet. Arguably, the "good ending" is the one that involves the survivors accepting blame for what humankind has done to their home planet and abandoning their hopes for asylum.

### 5.2 Integrating Environmental Design into Existing Curricula

In many cases, university departments may be reluctant or unable to create dedicated courses (modules) on environmental game studies or environmental design. This means that the choice of course structure, content, and teaching methods will often be dictated by the existing course framework. At Cologne Game Lab, we made use of the Collaborative Projects course, a 6-week long interdisciplinary group project with the aim to design a short playable game. This time frame is enough to produce an advanced prototype but does not afford sufficient time to familiarize oneself in-depth with a specific topic connected to environmental issues and conceive a design tailored to that topic. Using a preceding course with more of a theoretical focus would be a good way to prepare for the practical game design taking place in Collaborative Projects.

An alternative format–one employed in NordEcoJam–was a much shorter, 5-day game jam. This, too, would have benefited from pre-jam activities aimed to familiarize participants with existing environmental issues and encourage them to do independent research for their future projects.

A related challenge we encountered in NordEcoJam was that the partner universities' curricula and study regulations were not always compatible. This made the recognition of students' contributions in the game jam as part of their regular studies a challenge, since expectations and levels of administrative flexibility differed from school to school. While the benefits of inter-university collaboration far outweigh the costs of implementing it, navigating existing institutional structures requires careful planning and effort.

### 5.3 Finding Common Ground in Interdisciplinary, Cross-sectoral Collaboration

Meaningfully addressing environmental issues in game design education requires cooperation between many parties: game researchers, environmental scientists, game industry practitioners, and policy experts, to name a few. While necessary and ultimately rewarding, the challenge with such cooperation is that different parties have a different set of starting assumptions, goals, and often frame issues differently. An environmental scientist's explanation of ecosystem processes may prove too technical for other stakeholders (including students) to fully grasp without proper preparation; conversely, their expectations for student games in terms of complexity and representational fidelity may not be realistic. A game industry representative approaching the issue from a business perspective may not be enthusiastic about the concept of de-growth–scaling down consumption and production rather than incrementally increasing it as capitalist societies have been–even as many environmental scholars and activists advocate for its key role in environmental regeneration [16].

## 6 CONCLUSION

As humanity is facing a deepening climate crisis, educators across spheres and disciplines "have a responsibility to create inclusive environmental and sustainability educational approaches that are enabling, emotionally supportive, engaging, and praxis-oriented" [21]. Game design education is certainly no exception. It is imperative that future game professionals are not only aware of pressing environmental issues and the game industry's impact on them but also have enough knowledge, skills, and motivation to both advocate for systemic change and embody it in their work.

Our preliminary observations from the two case studies above are intended as starting points for discussion on possible approaches to environmentally conscious game design education, outlining both emerging potentialities and ongoing challenges.

Perhaps the biggest takeaway from our experience speaks to the need for educators to work together and to collaborate with industry representatives and associations, environmental scientists, and local authorities. This involves making an effort to locate and learn from existing knowledge in the field, as well as making the results of your own work available for others to use. Doing so can contribute to a burgeoning knowledge community whose shared goal is to harness videogames' dangers and potentialities in transitioning to a greener world.

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